

Ziyi LIU

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Availability: Full-time RA from **Apr/May 2026** (remote/on-site) for **6-12 months**.

Expected graduation **Dec 2026**. PhD applications planned for **2027**.

Research Interests: Computer vision; World Models; multimodal representations; Reliable AI.

EDUCATION

Master in Data and AI (GPA: 16/20)

Sept 2024 ~ Dec 2026

Institut Polytechnique de Paris (Télécom Paris / ENSTA / École Polytechnique)

Paris, France

Bachelor of Engineering in Intelligence & Robotics (3.0/4.0) Minor in Computer Science

Sept 2020 ~ Jul 2024

East China University of Science and Technology (ECUST)

Shanghai, China

- › ECUST **Outstanding Graduate** Scholarship (Top 25%, June 2024)
- › Huawei "Intelligence Base" Outstanding Student **Scholarship** (Nov 2021)

PATENTS AND PUBLICATIONS

- › **Z. Liu**, et al. FakeParts: A New Family of AI-Generated DeepFakes. Submitted to **CVPR 2026**.
- › Z. Zhang, **Z. Liu**, R. Bo. CAMA: A Culturally Adaptive Multi-Agent Framework for Postpartum Depression Support in Multilingual and Low-Resource Settings. **LM4UC @ AAAI 2025**.
- › Z. Long, et al., "Method and System for Predicting Wildfire Spread," Patent CN117077372A, Nov. 17, 2023.

SKILLS

- › **Languages:** English (IELTS 7.0), French (A2), Mandarin Chinese (Native), Cantonese (Fluent)
- › **Coding & Tech:** Python, C++, $\LaTeX$, Rust, HTML/CSS/JS/TS, Docker, Nginx, ROS, Git, Linux
- › Shanghai College Information Technology Level Examination: **Big Data and Cloud Computing**

RESEARCH EXPERIENCE

Research Assistant in WorldModel Bench

Jan 2026 ~ now

INRIA, supervised by **Dr. WANG Xi**

Paris, France

Research Assistant in Covariate Shift

Nov 2025 ~ now

ENSTA Paris U2IS, supervised by **Dr. Gianni Franchi**

Paris, France

- › **Development:** Diffusion-based methods for **covariate shift** detection in image tasks.
- › **Benchmarking:** Evaluation baselines on multiple **ID vs shift** image datasets, and defining the near- or far-Covariate Shift.

Research Intern in Deepfake Bench 🤖 FakeParts

Jun 2025 ~ Nov 2025

Hi! PARIS Lab, supervised by **Prof. Vicky Kalogeiton**

Paris, France

- › **Data Generation:** Built an automated video generation pipeline using **state-of-the-art generative models**, producing a ~**90K** benchmark dataset of **partially manipulated** high-quality videos.
- › **Benchmarking:** Conducted systematic benchmarking of **four model families** to validate the dataset and evaluate detection performance.

Multimodal Uncertainty Survey

Nov 2024 ~ Apr 2025

ENSTA Paris U2IS, supervised by **Dr. Gianni Franchi**

Paris, France

- › **Survey:** Authored a comprehensive survey on **multimodal model uncertainty**, synthesising insights from **over 80** recent works to construct a new taxonomy and identify emerging research trends.

Research Assistant in vSLAM

Jun 2023 ~ Nov 2023

ECUST Mobile Robot Lab, supervised by **Dr. Shuang Liu**

Shanghai, China

- › **Development:** Vision-based SLAM algorithms for indoor autonomous, implemented in **ROS**.
- › **Survey:** Feature extraction and matching methods: both **hand-crafted** and **learned** approaches.

SELECTED PROJECTS

Local Multimodal RAG System 🌀 MobileRAG

Aug 2025 ~ Current

- › **Automatic Incremental Updates:** Auto-differential embedding refreshing with **real-time document sync** for seamless updates of multimodal sources without full re-indexing.
- › **Multisource, Multimodal Retrieval:** Integrated heterogeneous data (Markdown, PDFs, images, audio, code) with unified vector stores and cross-modal encoders for **high-recall retrieval**.
- › **Edge-friendly Optimisation:** Employs **GPU quantisation**, **on-device caching**, batched ANN search, and memory-aware chunking to minimise latency and footprint.

Full-stack 3D Satellite Data Visualization 🌀 SATNET

Oct 2024 ~ Dec 2024

- › **3D Viz:** Built an interactive **satellite trajectory visualization** with **Cesium.js** and geospatial data processing using **d3.js**.
- › **Deployment:** Deployed the application on an **Ubuntu** server using **Vite.js**, **Nginx**, and **Docker**.

Personal Website 🌀 ZiyiLiu.top

Oct 2021 ~ Current

- › **Full-Stack Web:** Designed a **knowledge & portfolio platform** with **Express.js + TS** backend and **React + Vite + TS** frontend.
- › **Deployment:** Deployed across **multiple servers** with an **Nginx** reverse proxy and **cross-server Redis storage pool**, implementing secure load balancing, caching, and fault-tolerant data access.

OTHER EXPERIENCE

PFS Residual Value Assistant Intern

Jan 2024 ~ Jul 2024

Porsche Financial Leasing Ltd.

Shanghai, China

- › **Quantitative Modelling:** Built **Python and R** models to analyse and predict financial leasing product return rates, with back-testing for performance validation.
- › **Risk & Infrastructure:** Documented and optimised **credit strategy** and **fraud detection pipelines**, and proposed **data disaster tolerance solutions** to protect critical assets.

Back-end Developer Intern

Dec 2022 ~ Feb 2023

Bank of Communication Financial Technology Co., Ltd.

Shanghai, China

- › **HRM Platform:** Built a new **low-code Human Resource Management platform**, designing the **database schema** and automating the **end-to-end employee workflow**.
- › **AI & DevOps:** Implemented **OCR** and **face recognition** back-end functions **OpenAPI**, designed and executed **100+ test cases** for supply chain and debt management systems.

Development and Testing Intern

Apr 2022 ~ Sept 2022

Institute of Software, Chinese Academy of Sciences (ISCAS)

Remote

- › **System Integration:** Tested and packaged legacy software (**Autoconf-2.13**, **Pijul**) in the **openEuler** environment to ensure compatibility and functionality.
- › **Security Tooling:** Developed **rsensleak**, a **Rust**-based tool to detect and flag sensitive information leaks such as passwords and API keys in source code.

EXTRACURRICULAR ACTIVITIES

Head of Department, Media and Communication Sector

Sept 2020 ~ Aug 2022

Youth League Committee, ECUST

Shanghai, China

- › Managed the daily operations of a 20-member group and oversaw the college's WeChat official account "Huali Xinyuan," which has over 7,000 active users; awarded "4-star member media" by the Shanghai University Media Union in 2020 and 2021.
- › Organized college-wide orientation events for 300+ participants. Hosted two "Geek Festivals" in collaboration with other colleges and companies (Dell, Huawei), promoting information technology in creative and practical ways.